FLOODGAUGE · REGULATORY REGISTER

FloodGauge regulatory and standards register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/floodgauge		
Source	docs/07-regulatory.md @ dbb9a87		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	Initial draft for the kit 1.9.1 upgrade

FloodGauge regulatory and standards register

Which standards and regulations touch FloodGauge, where, and what is needed. Not legal advice; anything marked "to assess" needs a qualified reviewer before any product claim. Method: STANDARDS section 23. Verdict: the radio and public-space items decide whether a pilot can start; no standard here has been tested against the design, and every claim of conformity waits for the module certificates and a local review.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	47 CFR 15.255, operation within 57 to 71 GHz	United States	Unlicensed intentional radiators in the 60 GHz band, including radar	The 60 GHz street radar head of option A; the module maker's certification would be relied on, and its use as a fixed outdoor radar must be checked	Design lead	To assess
G2	ETSI EN 305 550 series, radiodetermination equipment 40 to 246 GHz, under the Radio Equipment Directive 2014/53/EU	European Union	Short-range radar in the 40 to 246 GHz range	Same radar head in the EU; scope for a fixed outdoor level radar is not confirmed	Design lead	To assess
G3	ETSI EN 300 220 (short range devices, 25 to 1,000 MHz), EU868 1 % duty cycle	European Union	LoRaWAN uplink timing and power	FieldNode radio; the event-mode interval keeps the sub-band duty cycle at or below 1 % (FLG-CAL-001 E1); owned by FieldNode	FieldNode owner	Met by design (CAL), certificate in the FieldNode repo
G4	The Things Network fair use policy, 30 s uplink airtime per day	Network rule	Airtime on the community network	Normal mode uses 23.7 s/day at SF9 (PR-007)	Design lead	Applies; at risk at SF10 and slower
G5	47 CFR Part 15.247 (902 to 928 MHz)	United States	Unlicensed spread-spectrum radio	FieldNode LoRaWAN radio at US pilot sites; the module's grant is relied on	FieldNode owner	To assess
G6	IEC 60529 ingress protection codes (IP67, IP68)	International	Dust and water ingress ratings	Drain head and street head; PR-009 asks IP68 at 2 m for 72 h and the transducer face is IP67 only	Design lead	At risk (CAL K1); immersion test at TRL 4
G7	Regulation (EU) 2016/679 (GDPR)	European Union	Personal data	Levels, status and battery only; no camera or microphone (PR-011); the device holds no personal data	Design lead	Does not apply as designed
G8	Directive 2011/65/EU (RoHS) and Directive 2012/19/EU (WEEE)	European Union	Hazardous substances and end-of-life of electronics	Would apply if units are sold in the EU; not for a self-built pilot	Design lead	Certification needed later
G9	Local road authority permission for a pole attachment, an arm over the gutter and anchors in the curb and sidewalk	Pilot city	Right to attach and to work in the public way	Arm at 4.6 m over the gutter (option A), stand pipe on the curb face (options B and C), eight anchors	Pilot partner	To assess

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G10	Local vehicle height limit and clearance rule for arms over the roadway	Pilot city	Minimum clearance above the road	The lens at 4.6 m clears a 4.3 m vehicle plus 0.3 m (CAL A7); the limit is typically 4.0 to 4.3 m and must be confirmed locally	Pilot partner	To assess
G11	Local rules on work at a catch basin or storm drain and on confined spaces	Pilot city	Entry and work in drains	The design never enters the basin: work is from the surface with the grate lifted (PR-012)	Pilot partner	Met by design; permit to assess
G12	UN 38.3 transport tests for lithium cells	International	Shipping the LiFePO4 cell	The cell is part of FieldNode; shipping a built unit is not planned at TRL 3	FieldNode owner	Certification needed later

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

Checked 2026-10-31 unless stated.

- Legal Information Institute. "47 CFR 15.255, Operation within the band 57 to 71 GHz." https://www.law.cornell.edu/cfr/text/47/15.255.
- ETSI. "EN 305 550-1, Short Range Devices (SRD), radio equipment to be used in the 40 GHz to 246 GHz frequency range, part 1." https://www.etsi.org/deliver/etsi_EN/305500_305599/30555001/01.01.01_60/en_30555001v010101p.pdf.
- The Things Network. "Fair Use Policy explained." https://www.thethingsnetwork.org/forum/t/fair-use-policy-explained/1300.
- The Radio Equipment Directive 2014/53/EU, ETSI EN 300 220, 47 CFR 15.247, IEC 60529, Regulation (EU) 2016/679, Directives 2011/65/EU and 2012/19/EU and UN 38.3 are named from general knowledge and were not opened in this session; each needs its text read before any claim (hence "to assess").
- The shared FieldNode standards (radio certification, battery) are researched once in the FieldNode repo and relied on here.